

LOW ACRYLONITRILE NBR PATENT LANDSCAPE REPORT

1. ABSTRACT

This report reviews patent evidence relevant to low acrylonitrile nitrile butadiene rubber (NBR), with emphasis on how acrylonitrile content is controlled during polymerization to obtain grades typically below about 20 wt% ACN. The single primary patent in the manifest, CN119978233A, discloses a low-temperature emulsion polymerization route for nitrile rubber in which butadiene and acrylonitrile are copolymerized with a reactive anti-aging agent containing amide functionality. The process uses staged or continuous acrylonitrile addition, a composite emulsifier system, a molecular weight regulator, and termination at controlled conversion to manage composition drift and final polymer properties. The patent explicitly targets a combined acrylonitrile content of 17-20%, placing it squarely in the low-ACN NBR domain.

The disclosed formulation trends are consistent with industrial low-ACN NBR practice: low-temperature emulsion polymerization, controlled monomer feed, and use of a mixed emulsifier package based on disproportionated rosin soap, linear alkylbenzene sulfonic acid, and naphthalene sulfonate formaldehyde condensate. The initiator system is based on dicumyl peroxide with a reducing agent/chelating agent package, and tertiary dodecyl mercaptan is used as a molecular weight regulator. The patent further emphasizes adding acrylonitrile in multiple portions, with a preferred four-stage feed tied to conversion milestones, to stabilize composition and improve heat-oxidative aging resistance.

Overall, the evidence indicates that low acrylonitrile NBR synthesis is best achieved by careful feed scheduling rather than simply lowering the initial monomer charge. The patent suggests that a butadiene/acrylonitrile ratio of 80-83/17-20 parts, polymerization at 5-15°C, and termination near 70-76% conversion are effective conditions for producing low-ACN NBR with improved thermal-oxidative durability for oil-resistant sealing applications.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: CN119978233A
- Patent Title: A method for preparing nitrile rubber with improved heat-oxidative aging ...
- Assignee: Not disclosed

- Jurisdiction: CN
- Publication Year: 2025
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses low-temperature emulsion copolymerization of butadiene and acrylonitrile to prepare nitrile rubber, with a reactive anti-aging agent containing amide groups incorporated as a functional monomer. Acrylonitrile is added in batches or continuously, preferably in more than three additions, and the latex is terminated, coagulated, washed, and dried after controlled conversion. - Vacuum degree: -0.1 MPa — Example 1 - Deionized water: 215 parts — Example 1 - Butadiene: 80 parts — Example 1 - Acrylonitrile: 11 parts — Example 1 - Reactive anti-aging monomer (N-(4-(anilino)phenyl)methacrylamide): 2.3 parts — Example 1 - Composite emulsifier: 2.8 parts — Example 1 - Disproportionated potassium abietate soap: 2.8 parts — Example 1 - Dodecylbenzenesulfonic acid: 1.1 parts — Example 1 - Naphthalene sulfonic acid formaldehyde condensate sodium salt: 0.38 parts — Example 1 - Sodium ethylenediamine tetraacetate: 0.11 parts — Example 1 - Disodium ethylenediamine tetraacetate: 0.04 parts — Example 1 - Tertiary dodecyl mercaptan: 0.6 parts — Example 1 - Reaction temperature: 7 ■ — Example 1 - Dicumyl hydroperoxide: 0.2 part — Example 1 - Secondary acrylonitrile addition at conversion: 30-35% — Example 1 - Secondary acrylonitrile: 3 parts — Example 1 - Tertiary acrylonitrile addition at conversion: 45-55% — Example 1 - Tertiary acrylonitrile: 3 parts — Example 1 - Quaternary acrylonitrile addition at conversion: 60-65% — Example 1 - Quaternary acrylonitrile: 3 parts — Example 1 - Sodium nitrite terminator: 0.07 part — Example 1 - Terminator addition at conversion: 73% — Example 1 - Butadiene content: 80-83 parts — Claim 2 - Acrylonitrile content: 17-20 parts — Claim 2 - Polymerization temperature: 5-15 ■ — Description - Preferred polymerization temperature: 8-12 ■ — Description - Disproportionated rosin acid soap: 2.0-4.0 parts — Description - Preferred disproportionated rosin acid soap: 2.5-4.0 parts — Description - C10-C13 linear alkylbenzene sulfonic acid: 1.0-2.0 parts — Description - Preferred C10-C13 linear alkylbenzene sulfonic acid: 1.0-1.5 parts — Description - Naphthalene sulfonic acid formaldehyde condensate sodium salt: 0.2-0.5 part — Description - Preferred naphthalene sulfonic acid formaldehyde condensate sodium salt: 0.3-0.5 part — Description - Dicumyl peroxide: 0.1-0.3 part — Description - Sodium ferric ethylenediamine tetraacetate: 0.01-0.15 part — Description - Disodium ethylenediamine tetraacetate: 0.01-0.05 part — Description - Tertiary dodecyl mercaptan: 0.3-0.8 part — Description - Preferred tertiary dodecyl mercaptan: 0.5-0.8 part — Description - Terminator: 0.05-0.15 part — Description - Terminator addition at conversion: 70-76% — Description - Deionized water: 200-230 parts — Description - Reactive anti-aging agent with amide groups: 1.5-2.5 parts — Description - Composite emulsifier: 3.2-6.5 parts — Description - Preferred composite emulsifier: 3.8-6.0 parts — Description

Relevant Experimental Evidence - Example 1: Demonstrated low-temperature emulsion polymerization with staged acrylonitrile addition, yielding nitrile rubber after coagulation and drying. - Example 1: Showed a feed schedule where acrylonitrile was added at 30-35%, 45-55%, 60-65%,

and termination at 73% conversion. - Examples 2-8: Used the same preparation route as Example 1 with Table 1 variations. - Comparative Examples 1-8: Used the same route with Table 2 variations for performance comparison. - Examples 1-8: Performance testing showed the invention can produce nitrile rubber with combined acrylonitrile content of 17-20% and improved thermal-oxidative aging resistance.

Technical Relevance This patent is directly relevant because it targets low-ACN NBR in the 17-20% range and discloses a practical emulsion polymerization strategy to control acrylonitrile incorporation. Its staged feed, low-temperature operation, and emulsifier/initiator package are all central process levers for low acrylonitrile NBR synthesis.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The available evidence shows that low acrylonitrile NBR is best produced by low-temperature emulsion copolymerization of butadiene and acrylonitrile with tight feed control. For the disclosed primary patent, the most relevant synthesis window is 80-83 parts butadiene and 17-20 parts acrylonitrile, polymerized at 5-15°C, with acrylonitrile added in multiple stages and termination near 70-76% conversion. A mixed emulsifier system and mercaptan chain-transfer agent support stable latex formation and molecular weight control. These conditions are consistent with producing low-ACN NBR suitable for oil-resistant sealing applications with improved heat-oxidative aging resistance.

5. REFERENCES

- CN119978233A | A method for preparing nitrile rubber with improved heat-oxidative aging ... | Not disclosed | CN | 2025 | <https://patents.google.com/patent/CN119978233A/en>